Lone Star Math Prep

Mathematics

Practice Test 1 • 2026

This test has 60 multiple-choice problems and lasts 40 minutes. Calculators are allowed. Mark the one best answer for each problem. Scoring follows the contest: 6 points for each correct answer, 2 points off for each wrong answer, and nothing for a problem left blank, so guess only when you can rule out some choices.

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1. For a thin lens, $1/f = 1/d_o + 1/d_i$. An object is 57 cm from a lens of focal length 38 cm. Find the image distance d_i in centimeters.

(A) 95 cm (B) 114 cm (C) 133 cm (D) 152 cm (E) 171 cm

2. Resistors of 230 Ω, 120 Ω and R_3 ohms are connected in parallel, so the total resistance R satisfies $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$. If R = 55.73 Ω, what is R_3? (nearest whole number)

(A) 190 (B) 230 (C) 270 (D) 310 (E) 350

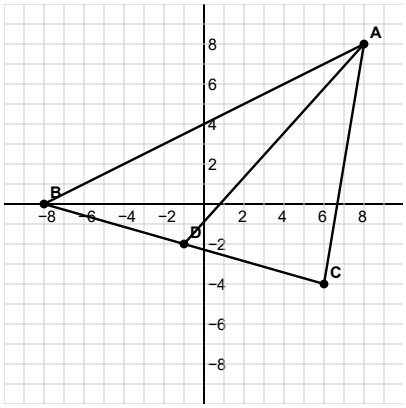
3. Luis's scores on 5 tests were 94, 93, 82, 92, 89. The final exam counts as two tests. What is the lowest whole-number score on the final exam that gives Luis an average of at least 92.2?

(A) 95 (B) 96 (C) 97 (D) 98 (E) 99

4–8. Triangle ABC has vertices A(8, 8), B(−8, 0), C(6, −4). Point D is the midpoint of side BC.

4. What is the y-intercept of the line through A and D?

(A) -1 (B) $-\frac{8}{9}$ (C) $-\frac{7}{9}$ (D) $-\frac{2}{3}$ (E) $-\frac{5}{9}$



5. What is the perimeter of △ADC? (nearest tenth)

(A) 32.9 (B) 33.1 (C) 33.3 (D) 33.5 (E) 33.7

6. What is the area of △ABD, in square units?

(A) 36 (B) 38 (C) 40 (D) 42 (E) 44

7. What is the measure of ∠BAC, in degrees? (nearest tenth)

(A) 53.7 (B) 54.0 (C) 54.3 (D) 54.6 (E) 54.9

8. What is the length of the longest median of △ADC? (nearest tenth)

(A) 11.7 (B) 11.9 (C) 12.1 (D) 12.3 (E) 12.5

9–11. Triangle ABC has AB = 65, BC = 36, and AC = 58. Triangle DEF is similar to triangle ABC, with D, E, F corresponding to A, B, C, and DF = 87. The bisector of ∠B meets AC at P.

9. Find the length of the median of triangle DEF drawn to DF. (nearest tenth)

(A) 65.1 (B) 65.3 (C) 65.5 (D) 65.7 (E) 65.9

10. Find AP. (nearest tenth)

(A) 36.9 (B) 37.1 (C) 37.3 (D) 37.5 (E) 37.7

11. Find the area of triangle PBC. (nearest whole number)

(A) 364 (B) 367 (C) 370 (D) 373 (E) 376

12–13. A community swimming pool is being filled.

12. A rectangular pool is 28 feet long, 29 feet wide and 6 feet deep. Use 1 cubic foot = 7.48052 gallons. How many gallons of water does it hold when full? (nearest whole number)

(A) 36,445 (B) 36,449 (C) 36,453 (D) 36,457 (E) 36,461

13. A pool can be filled by pipe A alone in 40 hours or by pipe B alone in 60 hours, and a drain can empty a full pool in 60 hours. Pipe A is opened at 8:00 AM, pipe B is opened at 10:00 AM, and the drain is opened at 8:00 PM, with all three left running. At what time is the pool full?

(A) 5:10 PM (B) 5:20 PM (C) 5:30 PM (D) 5:40 PM (E) 5:50 PM

14. Equilateral triangle RNK has side length 5 inches. Side RN is extended beyond N to point Z so that RZ = 12 inches. Find the area of triangle RKZ. (nearest tenth)

(A) 26.0 in² (B) 26.3 in² (C) 26.6 in² (D) 26.9 in² (E) 27.2 in²

15–16. Equilateral triangle WAT is inscribed in a circle of radius 4.

15. Find the length of minor arc WA. (nearest tenth)

(A) 8.1 (B) 8.4 (C) 8.7 (D) 9.0 (E) 9.3

16. Find the area of the region between chord WA and minor arc WA. (nearest tenth)

(A) 8.9 (B) 9.2 (C) 9.5 (D) 9.8 (E) 10.1

17. A circle is tangent to two adjacent sides of a rectangle and passes through a point P that is 12 units from one of these sides and 9 units from the other. Two circles fit this description; find the area of the smaller one. (nearest whole number)

(A) 122 (B) 125 (C) 128 (D) 131 (E) 134

18. From external point Z, tangent ZU has length 34. A secant from Z meets the circle at L and F, with L nearer Z, and ZL = 18. Find the length of ZF. (nearest tenth)

(A) 64.0 (B) 64.2 (C) 64.4 (D) 64.6 (E) 64.8

19–20. A chord of a circle with radius 26 lies 24 units from the center of the circle.

19. Find the length of the chord. (nearest tenth)

(A) 20.0 (B) 20.2 (C) 20.4 (D) 20.6 (E) 20.8

20. Find the area of the smaller region cut off by the chord. (nearest whole number)

(A) 11 (B) 15 (C) 19 (D) 23 (E) 27

21. Four consecutive even negative integers have this property: the product of the first and third is 1056 more than −15 times the fourth. What is the sum of the four integers?

(A) −160 (B) −156 (C) −152 (D) −148 (E) −144

22. In the arithmetic sequence 4, 13, 22, 31, 40, 49, … what is the sum of the 16th term and the 21st term?

(A) 321 (B) 323 (C) 325 (D) 327 (E) 329

23. Find the area of the circle with equation $x^2 + y^2 - 10x + 16y + 25 = 0$. (nearest whole number)

(A) 198 (B) 201 (C) 204 (D) 207 (E) 210

24. Let $f(x) = 7x^{\frac{7}{2}}$ for $x \geq 0$ and let g be the inverse of f. What is the smallest positive integer x for which $g(x) > 2$?

(A) 76 (B) 77 (C) 78 (D) 79 (E) 80

25. What is the square root of the 8th term of the geometric sequence 48, $28\frac{4}{5}$, $17\frac{7}{25}$, $10\frac{46}{125}$, $6\frac{138}{625}$, …? (nearest thousandth)

(A) 1.153 (B) 1.155 (C) 1.157 (D) 1.159 (E) 1.161

26–27. In a geometric sequence, the 3rd term is 72 and the 6th term is −243.

26. What is the 9th term? (nearest thousandth)

(A) 820.119 (B) 820.121 (C) 820.123 (D) 820.125 (E) 820.127

27. What is the sum of the first 9 terms? (nearest thousandth)

(A) 504.875 (B) 504.878 (C) 504.881 (D) 504.884 (E) 504.887

28. The temperature in a certain town varies sinusoidally over a 24-hour day, with a low of 46° at 3:00 AM and a high of 96° at 3:00 PM. For how many hours of the day is the temperature at or above 51°? (nearest tenth)

(A) 19.1 (B) 19.3 (C) 19.5 (D) 19.7 (E) 19.9

29. What is the range of the function $y = \text{Cos}^{-1}x$?

(A) $[-1, 1]$ (B) $(-1, 1)$ (C) $[-\pi, \pi]$ (D) $(0, \pi)$ (E) $[0, \pi]$

30. What is the sum of the x-coordinates of the two foci of the ellipse $25x^2 + 64y^2 + 150x - 512y - 351 = 0$? (nearest tenth)

(A) −6.4 (B) −6.2 (C) −6.0 (D) −5.8 (E) −5.6

31. An ellipse with center (−7, 5) has its axes parallel to the coordinate axes and is tangent to both the x-axis and the y-axis. What is its eccentricity? (nearest hundredth)

(A) 0.66 (B) 0.68 (C) 0.70 (D) 0.72 (E) 0.74

32. What is the period of the function $f(x) = -1 + 4\cos(5x + \frac{\pi}{2})$?

(A) $\frac{\pi}{5}$ (B) $\frac{2\pi}{5}$ (C) $\frac{4\pi}{5}$ (D) $\frac{6\pi}{5}$ (E) 2π

33. The circles $x^2 + y^2 = 17$ and $(x - 12)^2 + (y + 3)^2 = 68$ are externally tangent. Find the y-intercept of their common internal tangent line (the tangent line at the point where the circles touch). (nearest tenth)

(A) −17.2 (B) −17.0 (C) −16.8 (D) −16.6 (E) −16.4

34. Which of the following describes the graph of $4x^2 + 16xy + 16y^2 - 8x - 16y - 5 = 0$?

(A) ellipse (B) hyperbola (C) parabola (D) a line (E) two parallel lines

35. A ball is dropped from 13 feet and bounces to $\frac{2}{3}$ of its previous height each time. On which bounce is the height first below 2 feet?

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

36. A 200-gram mass hangs from a spring. It is pulled down 15 centimeters below its resting (equilibrium) position and released from rest at time t = 0, and it then oscillates up and down with a period of 0.8 seconds. How far above the equilibrium position is the mass at t = 1.83 seconds? (nearest tenth; a negative answer means below)

(A) 2.9 (B) 3.1 (C) 3.3 (D) 3.5 (E) 3.7

37. The polar curve $r = \cot^2\theta \csc\theta$ can be written in rectangular form as $y = h(x)$ for $x > 0$. What is $h(41)$? (nearest tenth)

(A) 10.7 (B) 11.1 (C) 11.5 (D) 11.9 (E) 12.3

38. Two circles with centers B and M and radii 7 and 8 are externally tangent. Tangent segments BG and BF are drawn from B to the larger circle, touching it at G and F. Find the area of the region inside quadrilateral BGMF that lies outside both circles. (nearest tenth)

(A) 9.4 (B) 9.6 (C) 9.8 (D) 10.0 (E) 10.2

39. Which of the following is a polar equation of the curve $xy = 12$?

(A) $r^2 = 12 \sin 2\theta$ (B) $r^2 = 24 \sec 2\theta$ (C) $r^2 = 24 \csc 2\theta$ (D) $r^2 = 48 \csc 2\theta$ (E) $r^2 = 24 \sin 2\theta$

40–41. Consider the graph of the equation $5x^2 + 3xy + 6y^2 - 8x - 5y - 8 = 0$.

40. What type of conic section is the graph?

(A) ellipse (B) parabola (C) hyperbola (D) line (E) 2 parallel lines

41. By rotating the axes through an acute angle θ, the xy-term of $5x^2 + 3xy + 6y^2 - 8x - 5y - 8 = 0$ can be eliminated. What is θ? (nearest tenth of a degree)

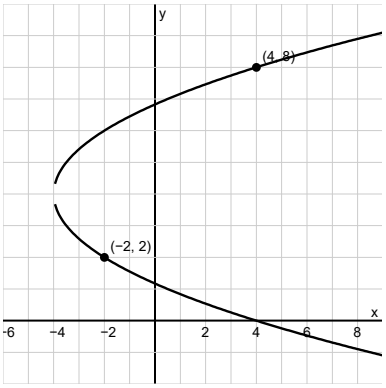
(A) 53.0° (B) 53.3° (C) 53.6° (D) 53.9° (E) 54.2°

42–45. The graph of a parabola with a horizontal axis is shown. Its vertex is V(−4, 4) and it passes through Q(−2, 2) and P(4, 8).

42. What is the equation of the directrix of the parabola?

(A) $x = -5.5$ (B) $x = -5$ (C) $x = -4.5$ (D) $x = -4$

(E) $x = -3.5$



43. What is the x-intercept of the line tangent to the parabola at Q(−2, 2)?

(A) 0 (B) 2 (C) 4 (D) 6 (E) 8

44. What is the area of the region bounded by the parabola and the chord PQ, where P(4, 8) and Q(−2, 2)? (nearest whole number)

(A) 18 (B) 19 (C) 20 (D) 21 (E) 22

45. What is the length of the arc of the parabola from its vertex V to P(4, 8)? (nearest tenth)

(A) 9.3 (B) 9.5 (C) 9.7 (D) 9.9 (E) 10.1

46–47. The population P of a colony of insects, with t in weeks, satisfies $dP/dt = 0.00005P(6000 - P)$, and $P = 600$ when $t = 0$.

46. At what time is the population growing most rapidly? (nearest tenth)

(A) 6.9 weeks (B) 7.1 weeks (C) 7.3 weeks (D) 7.5 weeks (E) 7.7 weeks

47. Find the population when $t = 18$ weeks. (nearest whole number)

(A) 5752 (B) 5759 (C) 5766 (D) 5773 (E) 5780

48. A particle moves along the parabola $y = 0.2x^2$ with horizontal velocity $dx/dt = 4 - 0.25t$ feet per second, where t is in seconds. When $t = 0$ the particle is at $x = 4$. What is the speed of the particle when $t = 2$? (nearest thousandth)

(A) 16.467 (B) 16.470 (C) 16.473 (D) 16.476 (E) 16.479

49. If $f''(x) = 12x^2 + 8$, $f(0) = 1$ and $f(2) = 25$, what is $f(-3)$?

(A) 127 (B) 130 (C) 133 (D) 136 (E) 139

50. The base of a solid is the part of the ellipse $49x^2 + 16y^2 = 784$ that lies in the third quadrant. Cross sections perpendicular to the x-axis are isosceles right triangles with a leg in the base. What is the volume of the solid? (nearest tenth)

(A) 64.7 (B) 64.9 (C) 65.1 (D) 65.3 (E) 65.5

51. What is the radius of convergence of the power series $\Sigma_{n=0}^{\infty} \frac{(n+2)(x+2)^n}{(3n+1)!}$?

(A) $\frac{1}{3}$ (B) $\frac{1}{2}$ (C) 4 (D) 9 (E) ∞

52–53. Scores on an exam are approximately normally distributed with mean 75 and standard deviation 12.

52. At what percentile is a score of 99? (nearest percentile)

(A) 90th (B) 92nd (C) 94th (D) 96th (E) 98th

53. What is the interquartile range of the scores? (nearest tenth)

(A) 16.2 (B) 16.6 (C) 17.0 (D) 17.4 (E) 17.8

54–55. A quantity is normally distributed with mean 1000 and standard deviation 50.

54. What proportion of the values exceed 1050? (nearest thousandth)

(A) 0.137 (B) 0.148 (C) 0.159 (D) 0.170 (E) 0.181

55. 9 independent values from this distribution are observed. Using $p = 0.1587$ as the probability that a single value exceeds 1050, what is the probability that at least 2 of the 9 values exceed 1050? (nearest thousandth)

(A) 0.418 (B) 0.430 (C) 0.442 (D) 0.454 (E) 0.466

56. A normal distribution has a standard deviation of 25. What is its interquartile range? (nearest tenth)

(A) 29.3 (B) 30.4 (C) 31.5 (D) 32.6 (E) 33.7

57–58. The table shows the miles a runner trained per week and the runner's 5K time in minutes. The least-squares regression line is fit to the data.

Miles per week	34	48	50	51	57	64
5K time	22.06	19.33	20.17	19.46	18.24	17.37

57. What is the residual for the point with miles = 48? (nearest thousandth)

(A) −0.529 (B) −0.527 (C) −0.525

(D) −0.523 (E) −0.521

58. Use the regression line to predict the 5K time when miles = 75. (nearest hundredth)

(A) 15.50 (B) 15.53 (C) 15.56 (D) 15.59 (E) 15.62

59–60. The table shows the number of visitors to a library on each day of one week.

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Visitors	61	56	57	52	51	60	60

59. What is the interquartile range of the daily counts? (Do not include the median in either half.)

(A) 8 (B) 10 (C) 12 (D) 14 (E) 16

60. A chi-square goodness-of-fit test is used to test whether visits are equally likely on all seven days. What is the value of the chi-square statistic? (nearest tenth)

(A) 1.3 (B) 1.5 (C) 1.7 (D) 1.9 (E) 2.1

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Mathematics Practice Test 1 • 2026
Answer Key

1. B	**21. B**	**41. E**
2. A	**22. B**	**42. C**
3. D	**23. B**	**43. B**
4. B	**24. E**	**44. A**
5. A	**25. D**	**45. A**
6. E	**26. D**	**46. C**
7. B	**27. A**	**47. C**
8. D	**28. A**	**48. D**
9. D	**29. E**	**49. B**
10. C	**30. C**	**50. D**
11. C	**31. C**	**51. E**
12. A	**32. B**	**52. E**
13. B	**33. B**	**53. A**
14. A	**34. E**	**54. C**
15. B	**35. D**	**55. B**
16. D	**36. D**	**56. E**
17. B	**37. D**	**57. B**
18. B	**38. A**	**58. E**
19. A	**39. C**	**59. A**
20. E	**40. A**	**60. C**

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Lone Star Math Prep

Mathematics

Practice Test 2 • 2026

This test has 60 multiple-choice problems and lasts 40 minutes. Calculators are allowed. Mark the one best answer for each problem. Scoring follows the contest: 6 points for each correct answer, 2 points off for each wrong answer, and nothing for a problem left blank, so guess only when you can rule out some choices.

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1. The kinetic energy of a moving object is $K = \frac{1}{2} m v^2$, with K in joules, m in kilograms, and v in meters per second. Find the mass when K = 271.8 J and v = 13.2 m/s. (nearest hundredth)

(A) 3.06 kg (B) 3.09 kg (C) 3.12 kg (D) 3.15 kg (E) 3.18 kg

2. The velocity of an object moving with constant acceleration a satisfies $v^2 = v_0^2 + 2a\Delta x$. Find a, in meters per second squared, when $v_0 = 35$ m/s, v = 69 m/s, and $\Delta x = 161.2$ m. (nearest hundredth)

(A) 10.64 (B) 10.75 (C) 10.86 (D) 10.97 (E) 11.08

3. A boarding stable charges a daily fee of $28.88, $3.75 per feeding, and $51.49 per hour for a riding lesson. A customer stays 18 days with 1 feeding and 60 minutes of riding lesson each day. What is the total charge?

(A) $1,509.72 (B) $1,510.83 (C) $1,511.94 (D) $1,513.05 (E) $1,514.16

4–5. The angles of a triangle measure $6x°$, $(4x - 10)°$, and $9x°$. The side opposite the largest angle is 20 units long.

4. Which of the following describes the triangle?

(A) acute (B) equilateral (C) isosceles (D) obtuse (E) right

5. Find the perimeter of the triangle. (nearest tenth)

(A) 47.1 (B) 47.3 (C) 47.5 (D) 47.7 (E) 47.9

6. Yusuf can frost 9 trays in 30 minutes and Rosa can frost 13 trays in 35 minutes. Yusuf starts at 6:00 AM and Rosa joins in at 6:30 AM. At what time are 92 trays finished? (nearest minute)

(A) 8:34 AM (B) 8:40 AM (C) 8:46 AM (D) 8:52 AM (E) 8:58 AM

7. A hiker walks 15 miles west, then 15 miles north, then 25 miles east, then 35 miles north. How many miles is she from her starting point? (nearest tenth)

(A) 51.0 (B) 51.2 (C) 51.4 (D) 51.6 (E) 51.8

8. A car's tires have a radius of 14 inches. When the car travels at 40 miles per hour, how many revolutions per minute does each tire make? (nearest whole number)

(A) 480 rpm (B) 486 rpm (C) 492 rpm (D) 498 rpm (E) 504 rpm

9. A right circular cone has a diameter of 22 and a slant height of 22. What is its volume? (nearest whole number)

(A) 2414 (B) 2422 (C) 2430 (D) 2438 (E) 2446

10. The graph of $f(x) = -x^2 + bx + c$ crosses the x-axis at x = 2 and x = 7. What is the maximum value of f(x)? (nearest tenth)

(A) 5.5 (B) 5.7 (C) 5.9 (D) 6.1 (E) 6.3

11. A rectangular pool is 78 feet long and 20 feet wide. A pump delivers 50 gallons per minute and takes 28.2 hours to fill the pool completely. Use 1 cubic foot = 7.48052 gallons. How deep is the pool, to the nearest inch?

(A) 7 ft 1 in (B) 7 ft 3 in (C) 7 ft 5 in (D) 7 ft 7 in (E) 7 ft 9 in

12. A right rectangular prism has a length of 9, a width of 25, and a volume of 4275. What is its total surface area? (nearest whole number)

(A) 1738 (B) 1742 (C) 1746 (D) 1750 (E) 1754

13–14. TG is a diameter of circle O with length 20. Point Z lies on the circle, and arc TZ measures 114°.

13. Find the area of sector ZOG. (nearest tenth)

(A) 57.2 (B) 57.6 (C) 58.0 (D) 58.4 (E) 58.8

14. Find the perimeter of triangle TOZ. (nearest tenth)

(A) 36.8 (B) 37.0 (C) 37.2 (D) 37.4 (E) 37.6

15. A right circular cone has a radius of 9 and a volume of 1272.3. What is its total surface area? (nearest whole number)

(A) 740 (B) 743 (C) 746 (D) 749 (E) 752

16. A rhombus has sides of length 23 cm and one angle of 30°. A circle has the same area as the rhombus. What is the circumference of the circle, in centimeters? (nearest whole number)

(A) 50 (B) 52 (C) 54 (D) 56 (E) 58

17–18. A circle with center O has an area of 363 square units. AC is a diameter and B is a point on the circle with $m\angle BAC = 30°$.

17. Find the area of triangle ACB. (nearest whole number)

(A) 100 (B) 103 (C) 106 (D) 109 (E) 112

18. Find the area of sector COB. (nearest tenth)

(A) 60.5 (B) 60.8 (C) 61.1 (D) 61.4 (E) 61.7

19–20. In right triangle ABC the right angle is at B, AB = 20, and AC = 30.7.

19. Find the area of triangle ABC. (nearest whole number)

(A) 221 (B) 224 (C) 227 (D) 230 (E) 233

20. Triangle DEF is similar to triangle ABC, with D, E, F corresponding to A, B, C, and EF = 6. The altitude from E meets DF at G. Find the perimeter of triangle DEG. (nearest tenth)

(A) 12.4 (B) 12.7 (C) 13.0 (D) 13.3 (E) 13.6

21–22. In rectangle ABCD, AB = 36. Point E lies on side BC with BE = 15, and $\angle AED = 90°$.

21. What is the area of triangle CDE? (nearest whole number)

(A) 1552 (B) 1555 (C) 1558 (D) 1561 (E) 1564

22. M is the midpoint of CD. What is the perimeter of triangle AEM? (nearest whole number)

(A) 224 (B) 227 (C) 230 (D) 233 (E) 236

23. The 7th term of an arithmetic sequence is 81 and the 11th term is 121. What is the sum of the first 20 terms?

(A) 2318 (B) 2320 (C) 2322 (D) 2324 (E) 2326

24–25. A swimming pool is a regular pentagonal prism: each side of its pentagonal floor is 25 feet and its depth is 3 feet. A fill pipe alone can fill the empty pool in 60 hours, and a drain alone can empty the full pool in 85 hours.

24. The pool is empty. The fill pipe is turned on, and after 12 hours the drain is opened as well. How many hours after the drain is opened is the pool full? (nearest whole number)

(A) 163 (B) 167 (C) 171 (D) 175 (E) 179

25. How many gallons of water does the pool hold when it is full? Use 1 cubic foot = 7.48 gallons. (nearest whole number)

(A) 24,130 (B) 24,134 (C) 24,138 (D) 24,142 (E) 24,146

26. A ball is dropped from a height of 8 feet and rebounds to 80% of its previous height after each bounce. How far has it traveled when it hits the ground for the 4th time? Give the answer in feet and inches. (nearest inch)

(A) 39 ft 0 in (B) 39 ft 3 in (C) 39 ft 6 in (D) 39 ft 9 in (E) 40 ft 0 in

27. The quadratic $6x^2 + bx + c$ has roots $\frac{3}{2}$ and $-\frac{4}{3}$. What is b + c?

(A) −15 (B) −13 (C) −11 (D) −9 (E) −7

28. In a geometric sequence the 1st term is 48 and the 6th term is −146.484375. What is the sum of the first 9 terms? (nearest thousandth)

(A) 180.178 (B) 180.279 (C) 180.380 (D) 180.481 (E) 180.582

29. What is the range of the function $y = \text{Tan}^{-1}x$?

(A) $(-\infty, \infty)$ (B) $[-\pi/2, \pi/2]$ (C) $(-\pi/2, \pi/2)$ (D) $(0, \pi)$ (E) $[0, \pi]$

30. What is the sum of the x-coordinates of the two foci of the ellipse $x^2 + 4y^2 - 14x + 16y + 1 = 0$? (nearest tenth)

(A) 13.2 (B) 13.4 (C) 13.6 (D) 13.8 (E) 14.0

31. Mia has made 70% of her attempted free throws this season. If she makes her next 42 in a row, her success rate will rise to 75%. How many attempts will she have in all after those 42?

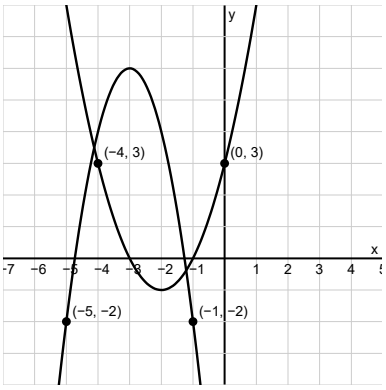
(A) 248 (B) 250 (C) 252 (D) 254 (E) 256

32–33. The graphs of two parabolas f and g with vertical axes are shown. The vertex of f is (−2, −1) and it passes through (0, 3); the vertex of g is (−3, 6) and it passes through (−1, −2).

32. What is the equation of the directrix of f?

(A) $y = -2\frac{1}{4}$ (B) $y = -2$ (C) $y = -1\frac{3}{4}$ (D) $y = -1\frac{1}{2}$

(E) $y = -1\frac{1}{4}$



33. What is the distance between the focus of f and the focus of g? (nearest tenth)

(A) 6.1 (B) 6.3 (C) 6.5 (D) 6.7 (E) 6.9

34. A radioactive sample has mass 4.932 g after 36 hours and 3.738 g after 48 hours. Assuming exponential decay, what was its original mass? (nearest hundredth)

(A) 11.00 g (B) 11.11 g (C) 11.22 g (D) 11.33 g (E) 11.44 g

35. A carrier sails 55° south of east at 18 mph. A jet leaves the carrier, flies 30° north of west at 560 mph, and then turns and flies straight back at the same speed to meet the moving carrier, landing exactly 5 hours after takeoff. How many miles from the carrier's starting point is the jet when it turns around? (nearest mile)

(A) 1353 (B) 1356 (C) 1359 (D) 1362 (E) 1365

36–37. Three ships leave the same port. Ship A leaves at 8:00 AM on a bearing of 90° at 24 mph, ship B leaves at 9:00 AM on a bearing of 180° at 22 mph, and ship C leaves at 10:00 AM on a bearing of 270° at 20 mph. Each keeps a constant course and speed.

36. At 2:00 PM, how many miles apart are ships B and C? (nearest whole number)

(A) 130 (B) 133 (C) 136 (D) 139 (E) 142

37. Ships A and C are on opposite bearings. At what time are they exactly 210 miles apart? (nearest minute)

(A) 1:29 PM (B) 1:32 PM (C) 1:35 PM (D) 1:38 PM (E) 1:41 PM

38. The polar graph of $r = \frac{8}{4 + 5\cos\theta}$ is a hyperbola. The point T(x, t) with x = 10 and t > 0 lies on its right branch, and the point R(x, r) with the same x lies on its asymptote that has positive slope. What is r − t? (nearest tenth)

(A) 0.6 (B) 0.8 (C) 1.0 (D) 1.2 (E) 1.4

39. A teacher hands out a study list of 26 problems and draws 4 of them at random, without replacement, for the exam. Jamal has studied 19 of the problems on the list. What is the probability that he can solve at least 3 of the 4 exam problems (he solves exactly the ones he has studied)? (nearest hundredth)

(A) 0.67 (B) 0.69 (C) 0.71 (D) 0.73 (E) 0.75

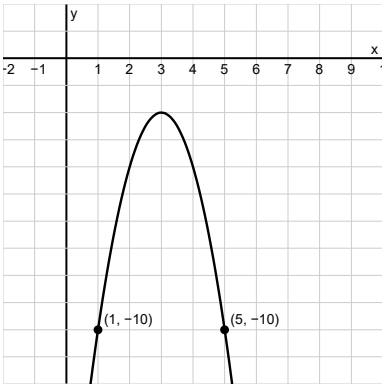
40. Five fair six-sided dice are rolled and the numbers showing are multiplied. What is the probability that the product is divisible by 4? (nearest ten-thousandth)

(A) 0.8643 (B) 0.8646 (C) 0.8649 (D) 0.8652 (E) 0.8655

41–42. The graph of a parabola with vertex (3, −2) is shown. It passes through (5, −10) and (1, −10).

41. What is the sum of the coordinates of the focus of the parabola?

(A) $\frac{5}{8}$ (B) $\frac{7}{8}$ (C) $1\frac{1}{8}$ (D) $1\frac{3}{8}$ (E) $1\frac{5}{8}$



42. If the parabola is the graph of y = f′(x) and f(1) = 1, what is f(6)?

(A) $-32\frac{2}{3}$ (B) $-32\frac{1}{3}$ (C) −32 (D) $-31\frac{2}{3}$ (E) $-31\frac{1}{3}$

43. The graph of $y = xe^x$ has a tangent line at the point where y = 4. Find the y-intercept of this tangent line. (nearest tenth)

(A) −5.1 (B) −4.8 (C) −4.5 (D) −4.2 (E) −3.9

44–45. Let R be the region in the first quadrant under the graph of $y = 22 - x^2$ from x = 2 to x = 4.

44. The area of R is approximated by the sum of the areas of four rectangles of equal width, using the right endpoint of each subinterval to find the height. What is the approximation?

(A) 22.00 (B) 22.25 (C) 22.50 (D) 22.75 (E) 23.00

45. What is the exact area of R?

(A) $25\frac{1}{6}$ (B) $25\frac{1}{4}$ (C) $25\frac{1}{3}$ (D) $25\frac{5}{12}$ (E) $25\frac{1}{2}$

46. Let g be a differentiable function and $f(x) = g(x^3 - 3x - 3)$. Then $f'(3) =$

(A) $3g'(15)$ (B) $15g'(15)$ (C) $18g'(15)$ (D) $24g'(15)$ (E) $48g'(15)$

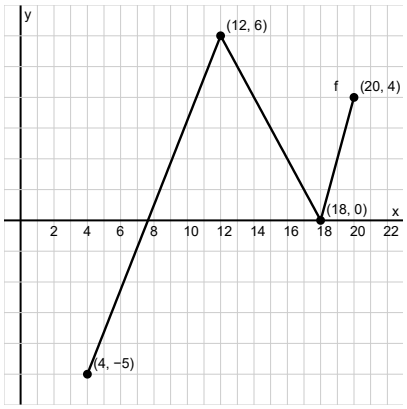
47. A point moves along the curve $x^2 + 3y^2 - xy = 45$. At the instant when $x = 3$ and $y > 0$, x is increasing at 3 units per second. How fast is y changing at that instant, in units per second? (nearest tenth)

(A) −0.7 (B) −0.5 (C) −0.3 (D) −0.1 (E) 0.1

48. A curve is given by the parametric equations $x = \ln(4t + 3)$, $y = 4t^2 + 4$. What is the value of d^2y/dx^2 at $t = \frac{1}{4}$? (nearest tenth)

(A) 9.4 (B) 9.7 (C) 10.0 (D) 10.3 (E) 10.6

49–50. The graph of the function h, made of line segments, is shown.



49. What is $\int_4^{20} h(x)\,dx$? (nearest whole number)

(A) 26 (B) 28 (C) 30 (D) 32 (E) 34

50. What is $\int_4^{20} h'(x)\,dx$? (nearest integer)

(A) 6 (B) 9 (C) 12 (D) 15 (E) 18

51. The correlation coefficient between a variable x and a variable y is r = 0.82. What is the correlation coefficient if the roles of x and y are reversed, so that y becomes the explanatory variable?

(A) −1.22 (B) −0.82 (C) 0.41 (D) 0.82 (E) 1.22

52. The probability distribution of a discrete random variable X is shown in the table. What is the expected value (mean) of X? (nearest hundredth)

x	3	4	5	6	7
P(x)	0.35	0.26	0.26	0.06	0.07

(A) 4.16 (B) 4.18 (C) 4.20 (D) 4.22 (E) 4.24

53. The weights of newborn lambs are approximately normally distributed with mean 4.2 kg and standard deviation 0.6 kg. What proportion of newborn lambs have a weight greater than 3.5 kg? (nearest hundredth)

(A) 0.76 (B) 0.79 (C) 0.82 (D) 0.85 (E) 0.88

54–55. The table shows the scores of nine students on a quiz worth 500 points.

54. What is the sum of the mean, the median and the mode of the scores?

Student	1	2	3	4	5	6	7	8	9
Score	322	357	315	461	293	457	294	291	315

(A) 971 (B) 973 (C) 975 (D) 977

(E) 979

55. By the 1.5·IQR rule, how many of the scores are outliers?

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

56–57. Heights of a group of students are approximately normally distributed with mean 71 inches and standard deviation 6.5 inches.

56. At what percentile is a person whose height is 6 ft 8 in? (nearest percentile)

(A) 86th (B) 88th (C) 90th (D) 92nd (E) 94th

57. What is the interquartile range of the heights, in inches? (nearest hundredth)

(A) 8.61 (B) 8.65 (C) 8.69 (D) 8.73 (E) 8.77

58. The scores on a placement exam are approximately normally distributed with a mean of 726 points. The 98th percentile of the scores is 970 points. What is the standard deviation of the distribution? (nearest tenth)

(A) 117.7 (B) 118.8 (C) 119.9 (D) 121.0 (E) 122.1

59. A screening test for a disease correctly identifies 91% of people who have it, and correctly gives a negative result for 97% of people who do not have it. 10% of a population has it. What percent of the population will test positive? (nearest tenth of a percent)

(A) 11.0% (B) 11.4% (C) 11.8% (D) 12.2% (E) 12.6%

60. The weights of newborn lambs are approximately normally distributed with mean 3.9 kg and standard deviation 0.7 kg. At what percentile is a newborn lamb with a weight of 3.4 kg? (nearest percentile)

(A) 20th (B) 22nd (C) 24th (D) 26th (E) 28th

Lone Star Math Prep
Mathematics Practice Test 2 • 2026
Answer Key

1. C	**21. B**	**41. B**
2. D	**22. C**	**42. B**
3. E	**23. B**	**43. B**
4. E	**24. A**	**44. B**
5. B	**25. A**	**45. C**
6. A	**26. B**	**46. D**
7. A	**27. B**	**47. C**
8. A	**28. B**	**48. C**
9. A	**29. C**	**49. A**
10. E	**30. E**	**50. B**
11. B	**31. C**	**51. D**
12. B	**32. E**	**52. E**
13. B	**33. D**	**53. E**
14. A	**34. D**	**54. C**
15. D	**35. C**	**55. A**
16. E	**36. C**	**56. D**
17. A	**37. E**	**57. E**
18. A	**38. C**	**58. B**
19. E	**39. C**	**59. C**
20. A	**40. B**	**60. C**

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Lone Star Math Prep

Mathematics

Practice Test 3 • 2026

This test has 60 multiple-choice problems and lasts 40 minutes. Calculators are allowed. Mark the one best answer for each problem. Scoring follows the contest: 6 points for each correct answer, 2 points off for each wrong answer, and nothing for a problem left blank, so guess only when you can rule out some choices.

DO NOT TURN THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO

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1. On a contest team, three students each answered all 55 problems and missed 5, 6, and 3 of them. Each correct answer scores 6 points and each wrong answer loses 2 points. What was the team's total score?

(A) 876 (B) 878 (C) 880 (D) 882 (E) 884

2. A lunch check lists a milkshake at \$4.89, an iced tea at \$2.02, a chicken quesadilla at \$10.99, and a lemonade at \$3.66. Sales tax of 7.75% is charged on the food, and a flat tip of \$4.00 is added. What is the total cost? (nearest cent)

(A) \$26.67 (B) \$26.81 (C) \$26.95 (D) \$27.09 (E) \$27.23

3. Line m passes through (1, 3) and (5, −5). Line n is parallel to m and passes through (−8, −6). What is the y-intercept of line n?

(A) −22 (B) −20 (C) −18 (D) −16 (E) −14

4–7. Triangle ABC has vertices A(8, −8), B(2, 6), C(−6, 4).

4. The point D(8, d) is such that AB ⊥ CD. What is d?

(A) 6 (B) 7 (C) 8 (D) 9 (E) 10

5. What is the perimeter of △ABC? (nearest tenth)

(A) 41.5 (B) 41.7 (C) 41.9 (D) 42.1 (E) 42.3

6. The point E(5, c) is such that AB ∥ CE. What is c? (nearest tenth)

(A) −22.5 (B) −22.3 (C) −22.1 (D) −21.9 (E) −21.7

7. What is the distance between the midpoints of AB and BC? (nearest tenth)

(A) 9.0 (B) 9.2 (C) 9.4 (D) 9.6 (E) 9.8

8. A car's tires have a radius of 18 inches. When the car travels at 50 miles per hour, how many revolutions per minute does each tire make? (nearest whole number)

(A) 461 rpm (B) 467 rpm (C) 473 rpm (D) 479 rpm (E) 485 rpm

9. A collection has pens, pencils, and markers. The number of pencils is 8 more than 2 times the number of pens. The number of markers is 5 more than 5 times the number of pencils. There are 300 items altogether. How many markers are there?

(A) 235 (B) 237 (C) 239 (D) 241 (E) 243

10. A jar holds only pennies and dimes. One thousand pennies have a mass of 2.5 kg and one thousand dimes have a mass of 2.268 kg. The coins in the jar are worth \$92.00 and have a total mass of 4.8144 kg. How many dimes are in the jar?

(A) 785 (B) 790 (C) 795 (D) 800 (E) 805

11. A chord of length 20 is drawn in a circle with radius 15. Find the length of the minor arc cut off by the chord. (nearest tenth)

(A) 20.7 (B) 21.0 (C) 21.3 (D) 21.6 (E) 21.9

12–13. Quadrilateral KNMW is inscribed in a circle. m∠WKN = 63°, m∠KNM = 60°, and arc MW (the arc not containing K or N) measures 49°.

12. Find m∠NMW.

(A) 116° (B) 117° (C) 118° (D) 119° (E) 120°

13. Find the measure of arc KN (the arc not containing M or W).

(A) 163° (B) 165° (C) 167° (D) 169° (E) 171°

14–15. NX is a diameter of circle O with length 30. Point C lies on the circle, and arc NC measures 108°.

14. Find the area of sector COX. (nearest tenth)

(A) 140.2 (B) 140.6 (C) 141.0 (D) 141.4 (E) 141.8

15. Find the perimeter of triangle NOC. (nearest tenth)

(A) 54.1 (B) 54.3 (C) 54.5 (D) 54.7 (E) 54.9

16. A regular octagon is inscribed in a circle of area 219. What is the area of the polygon? (nearest whole number)

(A) 194 (B) 197 (C) 200 (D) 203 (E) 206

17. In right triangle THY the right angle is at Y. The median from T to the midpoint of HY has length 65, and the median from H to the midpoint of TY has length 62. Find the perimeter of the triangle. (nearest whole number)

(A) 194 (B) 196 (C) 198 (D) 200 (E) 202

18–19. A circle with center O has an area of 606 square units. GL is a diameter and K is a point on the circle with $m\angle KGL = 30°$.

18. Find the area of triangle GLK. (nearest whole number)

(A) 161 (B) 164 (C) 167 (D) 170 (E) 173

19. Find the area of sector LOK. (nearest tenth)

(A) 100.7 (B) 101.0 (C) 101.3 (D) 101.6 (E) 101.9

20. A solid is made by placing a hemisphere on top of a right circular cylinder; they have the same radius, and the height of the cylinder is 4 times that radius. The total surface area of the solid (including the circular bottom) is 2610. What is the volume of the solid? (nearest whole number)

(A) 9611 (B) 9615 (C) 9619 (D) 9623 (E) 9627

21. An investment of \$105,000.00 gains 13% in its first year, gains 22% in its second year, and changes by −4.0% in its third year. What was the average annual percent increase over the three years? (nearest tenth)

(A) 9.0% (B) 9.2% (C) 9.4% (D) 9.6% (E) 9.8%

22. The vector $13i - j$ has magnitude 13.0. Which of the following is a direction angle θ for the vector, in degrees, measured from the positive x-axis? (nearest tenth)

(A) −184.4° (B) −175.6° (C) −4.4° (D) 4.4° (E) 175.6°

23. \$20,914.44 is deposited in an account paying 5.23% annual interest compounded monthly. In how many years will the balance have tripled? (nearest whole number)

(A) 17 yr (B) 18 yr (C) 19 yr (D) 20 yr (E) 21 yr

24. The inverse of a function f is $f^{-1}(x) = \frac{ax}{-3x - 2}$, where a is a constant. If $f^{-1}(1) = 1$, what is f(5)?

(A) −5 (B) −4 (C) −3 (D) −2 (E) −1

25–26. A regular octagon has sides of length 6.

25. What is the area of the polygon? (nearest whole number)

(A) 170 (B) 174 (C) 178 (D) 182 (E) 186

26. What is the area of the circle inscribed in the polygon? (nearest whole number)

(A) 165 (B) 169 (C) 173 (D) 177 (E) 181

27. Two boats leave the same dock. The first leaves at 10:00 AM on a bearing of 098°. The second leaves at 11:30 AM on a bearing of 259° and travels 6 miles per hour faster than the first. Each moves in a straight line at a constant speed, and at 3:30 PM the boats are 229.8 miles apart. What is the speed of the second boat, in miles per hour? (nearest whole number)

(A) 27 (B) 28 (C) 29 (D) 30 (E) 31

28. Two planes leave the same airport. The first plane leaves at 9:30 AM on a course of N 32° W at 160 miles per hour. The second plane leaves at 10:30 AM on a course of S 68° E at 218 miles per hour. Each travels in a straight line at a constant speed. How far apart are they at 12:30 PM? (nearest mile)

(A) 867 (B) 869 (C) 871 (D) 873 (E) 875

29–30. Let $f(x) = x^5 - 2x^4 + 4x^3 + dx^2 - 2x + 2$, where d is a constant and $f(-1) = -11$.

29. What is the value of d?

(A) −12 (B) −10 (C) −8 (D) −6 (E) −4

30. Using the value of d from the previous problem, how many real zeros does f have?

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

31. A hyperbola centered at the origin has asymptotes $y = \pm\frac{1}{2}x$ and a vertex at (0, 6). What is the distance between the foci? (nearest tenth)

(A) 26.6 (B) 26.8 (C) 27.0 (D) 27.2 (E) 27.4

32. The point (4, k) is equidistant from P(−6, −3) and Q(2, 3). What is k? (nearest tenth)

(A) −8.0 (B) −6.9 (C) −5.8 (D) −4.7 (E) −3.6

33. On a certain planet a day lasts 48 hours. The temperature varies sinusoidally over the day, with a low of 17° at time t = 0 and a high of 44° at t = 24 hours. For how many hours of each day is the temperature at or above 27°? (nearest tenth)

(A) 27.1 (B) 27.4 (C) 27.7 (D) 28.0 (E) 28.3

34. What is the perimeter of the triangle with vertices (6, 9, −4), (−8, 7, −3) and (−2, −5, −6)? (nearest tenth)

(A) 43.8 (B) 44.0 (C) 44.2 (D) 44.4 (E) 44.6

35. On Earth, a ball is thrown from a height of 78 feet with an initial speed of 60 feet per second at an angle of 38° above the horizontal. Its position t seconds later is $x = (60\cos 38°)t$, $y = 78 + (60\sin 38°)t - 16.087t^2$ (feet). How far has the ball traveled horizontally when it lands? (nearest foot)

(A) 172 (B) 175 (C) 178 (D) 181 (E) 184

36. The parametric equations $x = e^{t+3}$, $y = 3e^{2t}$ describe a curve that can be written as y = f(x). What is f(6)? (nearest tenth)

(A) 0.3 (B) 0.5 (C) 0.7 (D) 0.9 (E) 1.1

37. A club has 9 seniors and 11 juniors. It elects a president, a vice president and a treasurer, all different people, and the president must be a senior. In how many ways can the offices be filled?

(A) 990 (B) 3078 (C) 3249 (D) 3420 (E) 6840

38. How many petals does the rose $r = 14\cos 3\theta$ have?

(A) 3 (B) 4 (C) 5 (D) 7 (E) 12

39–40. The parabola $x^2 + 10x - 4y + 21 = 0$ and the line $x + 2y = -6$ are graphed in the plane.

39. The focus of the parabola is (p, q). What is $p + q$?

(A) -9 (B) -8 (C) -7 (D) -6 (E) -5

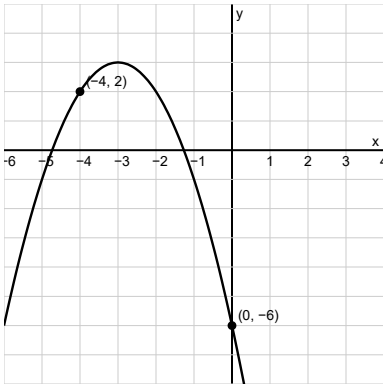
40. The parabola and the line intersect at two points. What is the distance between them? (nearest tenth)

(A) 3.1 (B) 3.3 (C) 3.5 (D) 3.7 (E) 3.9

41–43. The graph of a parabola with a vertical axis is shown. Its vertex is $(-3, 3)$, and it passes through $P(-4, 2)$ and $Q(0, -6)$.

41. What is the y-coordinate of the focus of the parabola?

(A) 2.75 (B) 3.00 (C) 3.25 (D) 3.50 (E) 3.75



42. What is the slope of the line tangent to the parabola at $Q(0, -6)$? (nearest tenth)

(A) -6.0 (B) -5.8 (C) -5.6 (D) -5.4 (E) -5.2

43. What is the area of the region bounded by the parabola and the line through $P(-4, 2)$ and $Q(0, -6)$? (nearest tenth)

(A) 10.5 (B) 10.7 (C) 10.9 (D) 11.1 (E) 11.3

44. A running track is made of a rectangle with a semicircle at each of two opposite ends. The track is 500 meters around. Among all such tracks, the rectangular region in the middle is made as large as possible. What is the radius of each semicircle? (nearest tenth)

(A) 39.4 m (B) 39.6 m (C) 39.8 m (D) 40.0 m (E) 40.2 m

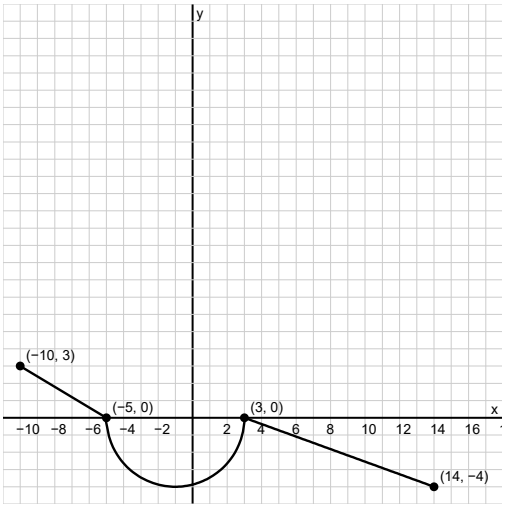
45. The function $f(x) = -x^3 - 3x^2 - 7x - 3$ has an inverse function g. Find $g'(7)$.

(A) $-\frac{1}{3}$ (B) $-\frac{1}{4}$ (C) $-\frac{1}{5}$ (D) $-\frac{1}{6}$ (E) $-\frac{1}{7}$

46–47. The graph of f′, the derivative of f, is shown for $-10 \le x \le 14$. It consists of two line segments and a semicircle. It is known that $f(-6) = 6$.

46. What is the absolute maximum value of f on the interval? (nearest whole number)

(A) 4 (B) 6 (C) 8 (D) 10 (E) 12



47. What is the absolute minimum value of f on the interval? (nearest hundredth)

(A) −40.87 (B) −40.83 (C) −40.79 (D) −40.75 (E) −40.71

48–49. A function y satisfies $dy/dx = 3x^2y$ with $y(0) = 3$.

48. Use Euler's method with two equal steps to approximate y(0.6). (nearest hundredth)

(A) 3.10 (B) 3.17 (C) 3.24 (D) 3.31 (E) 3.38

49. Solve the differential equation exactly and find y(0.6). (nearest hundredth)

(A) 3.66 (B) 3.69 (C) 3.72 (D) 3.75 (E) 3.78

50. The curve $x^2 + 4xy + 4y^2 = 16$ passes through the point with $x = 2$ and $y > 0$. Find dy/dx at that point. (nearest hundredth)

(A) −0.56 (B) −0.53 (C) −0.50 (D) −0.47 (E) −0.44

51. The table gives the joint probabilities for events A and B. Find P(A | B) as a decimal.

(A) 0.05 (B) $0.\overline{1}$ (C) 0.15 (D) $0.\overline{3}$ (E) 0.45

	B	not B
A	0.05	0.10
not A	0.40	0.45

52. For a function f, $f(2) = 2$, $f'(2) = -3$, $f''(2) = -8$ and $f'''(2) = -8$. Use the third-degree Taylor polynomial of f about $x = 2$ to approximate $f(2.1)$. (nearest hundredth)

(A) 1.63 (B) 1.66 (C) 1.69 (D) 1.72 (E) 1.75

53. A two-sided z test for a population proportion gives the test statistic $z = -1.86$. What is the P-value? (nearest thousandth)

(A) 0.039 (B) 0.045 (C) 0.051 (D) 0.057 (E) 0.063

54–55. The table shows the number of workers on a shift and the units produced. The least-squares regression line is fit to the data.

54. What is the slope of the least-squares regression line? (nearest tenth)

(A) 7.0 (B) 7.3 (C) 7.6 (D) 7.9 (E) 8.2

Workers	3	6	9	12	15	18
Units	101	111	142	163	175	206

55. What is the residual for the point with workers = 15? (nearest tenth)

(A) −6.9 (B) −6.7 (C) −6.5 (D) −6.3 (E) −6.1

56. Four swimmers' lap times, in seconds, are independent and normally distributed with the means and standard deviations in the table. What is the probability that the total of the four times is less than 108.8 seconds? (nearest hundredth)

Swimmer	Mean (sec)	SD (sec)
Ann	30.2	0.7
Bo	29.2	1.4
Cy	27.2	1.3
Di	24.1	1.5

(A) 0.23 (B) 0.26 (C) 0.29 (D) 0.32 (E) 0.35

57. The probability distribution of a discrete random variable X is shown in the table. What is the expected value (mean) of X? (nearest hundredth)

x	2	3	4	5	6
P(x)	0.09	0.12	0.19	0.41	0.19

(A) 4.45 (B) 4.47 (C) 4.49 (D) 4.51 (E) 4.53

58–59. A teacher recorded the weekday on which each of 190 late assignments was turned in; the counts are shown in the table. A chi-square goodness-of-fit test is used to test the claim that late assignments are equally likely on each of the 5 days.

58. What is the P-value of the chi-square goodness-of-fit test? (nearest thousandth)

Day	Mon	Tue	Wed	Thu	Fri
Count	28	46	27	47	42

(A) 0.040 (B) 0.046 (C) 0.052 (D) 0.058 (E) 0.064

59. What is the contribution of the Tue category to the chi-square statistic? (nearest hundredth)

(A) 1.46 (B) 1.68 (C) 1.90 (D) 2.12 (E) 2.34

60. Scores on a 60-problem contest are approximately normal with mean 126 and standard deviation 60. The champion's score was 2.5 standard deviations above the mean (z = 2.5). The contest awards 6 points for each correct answer, subtracts 2 points for each wrong answer, and gives 0 for a problem left blank. The champion answered 50 of the 60 problems. How many of those answers were wrong?

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Lone Star Math Prep
Mathematics Practice Test 3 • 2026
Answer Key

1. B	**21. E**	**41. A**
2. E	**22. C**	**42. A**
3. A	**23. E**	**43. B**
4. E	**24. E**	**44. C**
5. C	**25. B**	**45. E**
6. E	**26. A**	**46. B**
7. B	**27. B**	**47. B**
8. B	**28. C**	**48. C**
9. A	**29. C**	**49. C**
10. D	**30. C**	**50. C**
11. E	**31. B**	**51. B**
12. B	**32. A**	**52. B**
13. A	**33. D**	**53. E**
14. D	**34. C**	**54. A**
15. B	**35. A**	**55. D**
16. B	**36. A**	**56. A**
17. A	**37. B**	**57. C**
18. C	**38. A**	**58. A**
19. B	**39. E**	**59. B**
20. D	**40. E**	**60. D**

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Practice Test 1 — how each answer is found

#	Answer	Method
1	B 114 cm	$1/d_i = 1/f - 1/d_o$.
2	A 190	Subtract the known reciprocals from 1/R, then take the reciprocal.
3	D 98	$(450 + 2F)/7 \geq 92.2$, then round F up.
4	B $-\frac{8}{9}$	Slope from A to D, then $b = y - m \cdot x$ at D.
5	A 32.9	Add the three side lengths found with the distance formula.
6	E 44	Shoelace formula: half the absolute value of the cross-product sum.
7	B 54.0	Dot product of the two side vectors divided by the product of their lengths, then inverse cosine.
8	D 12.3	Compute all three medians (vertex to midpoint of the opposite side); take the largest.
9	D 65.7	Median to AC: $\frac{1}{2}\sqrt{(2a^2 + 2c^2 - b^2)}$; scale by DF/AC.
10	C 37.3	Angle bisector theorem: AP/PC = AB/BC.
11	C 370	Heron's formula for ABC; triangle PBC has PC/AC of the area.
12	A 36,445	V = lwd in cubic feet, times 7.48052.
13	B 5:20 PM	Add the fraction of the pool filled in each stage, then divide the rest by the net rate.
14	A 26.0 in²	Area = $\frac{1}{2} \cdot RK \cdot RZ \cdot \sin 60°$.
15	B 8.4	Each side cuts off a 120° arc: one third of the circumference.
16	D 9.8	One third of (circle area − triangle area).
17	B 125	Center (r, r): $(r - a)^2 + (r - b)^2 = r^2$, smaller root $r = (a + b) - \sqrt{(2ab)}$; area πr^2.
18	B 64.2	$ZU^2 = ZL \cdot ZF$, so $ZF = 34^2 \div 18$.
19	A 20.0	Half-chord $= \sqrt{(r^2 - d^2)}$.
20	E 27	Sector minus triangle: $\frac{1}{2}r^2(\theta - \sin\theta)$, $\theta = 2\cos^{-1}(d/r)$.
21	B −156	$x(x + 4) = 1056 + (-15)(x + 6)$ gives $x = -42$; the sum is $4x + 12$.
22	B 323	Common difference d, then $a_n = a_1 + (n-1)d$ for each term.
23	B 201	Complete the square: $(x - h)^2 + (y - k)^2 = r^2$; the area is πr^2.
24	E 80	$g(x) > t$ exactly when $x > f(t)$; take the next integer above f(t).
25	D 1.159	Common ratio from consecutive terms, $a_n = a_1 r^{n-1}$, then the root.
26	D 820.125	$r^3 = a_k/a_j$ fixes r (negative here); then $a_n = a_1 r^{n-1}$.
27	A 504.875	$S_n = a_1(1 - r^n)/(1 - r)$.
28	A 19.1	$T = \text{mid} - \text{amp} \cdot \cos(2\pi t/P)$; $T \geq \text{thr}$ when cos is small enough: time $= (P/\pi) \cdot \arccos((\text{thr} - \text{mid})/\text{amp})$.
29	E $[0, \pi]$	$\cos^{-1}$ takes the values between its limiting angles; the endpoints are included only if they are attained.
30	C −6.0	Complete the squares; the foci lie on the major axis, symmetric about the center.
31	C 0.70	The semi-axes are $\lvert h\rvert$ and $\lvert k\rvert$; $e = \sqrt{(a^2 - b^2)}/a$ with a the larger.
32	B $\frac{2\pi}{5}$	Period of the parent (2π for sin, cos, sec, csc; π for tan, cot) divided by the coefficient of x.
33	B −17.0	The touching point divides the line of centers in the ratio of the radii; the tangent is perpendicular there.
34	E two parallel lines	$B^2 - 4AC = 0$ suggests a parabola, but the left side factors into two parallel linear factors.
35	D 5	Solve $h \cdot r^n <$ threshold with logarithms and round the exponent up.
36	D 3.5	$y = -A \cdot \cos(2\pi t/P)$.

#	Answer	Method
37	D 11.9	Here x = cot^3θ and y = cot^2θ, so y = x^($\frac{2}{3}$).
38	A 9.4	Kite area = $r_2 \cdot \sqrt{(d^2 - r_2^2)}$ (two right triangles) minus the two circular sectors at B and M.
39	C $r^2 = 24 \csc 2\theta$	Substitute x = r cos θ, y = r sin θ, simplify, and use the double-angle identity.
40	A ellipse	Discriminant $B^2 - 4AC$: negative ellipse, zero parabola, positive hyperbola.
41	E 54.2°	tan 2θ = B/(A − C).
42	C x = −4.5	p = 1/(4a) from the points; the directrix is |p| behind the vertex.
43	B 2	dx/dy = 2a(y − k); the tangent line $x = x_0 + (dx/dy)(y - y_0)$ at y = 0.
44	A 18	Integrate in y: area = $\|a\|\Delta y^3/6$.
45	A 9.3	Arc length $\int \sqrt{(1 + (dx/dy)^2)}\, dy$ from y = k to the y of P.
46	C 7.3 weeks	Growth is fastest at P = M/2; solve the logistic solution for t.
47	C 5766	$P = M/(1 + Ae^{-kMt})$, $A = (M - P_0)/P_0$.
48	D 16.476	$x = x_0 + qt + pt^2/2$; speed = $\|x'\|\sqrt{(1 + (2ax)^2)}$.
49	B 130	Integrate twice; use the two values to find the constants, then evaluate.
50	D 65.3	Base length $L = b\sqrt{(1 - x^2/a^2)}$; triangle area $L^2/2$ (leg) or $L^2/4$ (hypotenuse); integrate over one half of the ellipse.
51	E ∞	The factorial grows faster than any power, so the series converges for every x.
52	E 98th	z = 2
53	A 16.2	IQR = 2(0.6745)σ
54	C 0.159	z = (1050 − 1000) ÷ 50 = 1.0; P(Z > 1.0)
55	B 0.430	Binomial P(X ≥ 2) with n = 9, p = 0.1587
56	E 33.7	IQR = 2(0.6745)σ
57	B −0.527	Residual = y − ŷ, with ŷ from the least-squares line (y = a + bx)
58	E 15.62	ŷ = a + b·x from the least-squares line
59	A 8	IQR = Q3 − Q1, quartiles being the medians of the lower and upper halves
60	C 1.7	$\chi^2 = \Sigma (O - E)^2 \div E$ with E = total ÷ 7

Practice Test 2 — how each answer is found

#	Answer	Method
1	C 3.12 kg	Solve the formula for the unknown and substitute.
2	D 10.97	$a = (v^2 - v_0^2)/(2\Delta x)$.
3	E $1,514.16	d × (daily fee + meals × price + hours × rate).
4	E right	Solve for x (the angles sum to 180°): the angles are 60°, 30°, 90°, all different, and the largest is equal to 90°.
5	B 47.3	Law of sines: each side is (given side)·sin(angle)/sin(largest angle); add the three sides.
6	A 8:34 AM	Work done alone first, then the rest at the combined rate.
7	A 51.0	Net east-west and north-south displacements, then the Pythagorean theorem.
8	A 480 rpm	Inches per minute = mph × 63,360 ÷ 60; divide by the tire circumference 2πr.
9	A 2414	$h = \sqrt{(l^2 - r^2)}$; $V = (\frac{1}{3})\pi r^2 h$.
10	E 6.3	The maximum is at the midpoint of the zeros.
11	B 7 ft 3 in	Gallons = rate × time; divide by 7.48052 for cubic feet, then by the floor area; convert to feet and inches.
12	B 1742	h = V ÷ (lw); TSA = 2(lw + lh + wh).
13	B 57.6	Sector ZOG has central angle 180 − 114 = 66°: area = $(\frac{66}{360})\pi \cdot 10^2$.
14	A 36.8	Two radii of 10 plus the chord 2·10·sin(57°).
15	D 749	$h = 3V \div (\pi r^2)$, slant $l = \sqrt{(r^2 + h^2)}$, then $TSA = \pi r^2 + \pi rl$.
16	E 58	Rhombus area = $s^2 \sin\theta$; then $r = \sqrt{(A/\pi)}$ and $C = 2\pi r$.
17	A 100	Right angle at B: legs 2r·cos 30° and 2r·sin 30°, so area = $r^2 \sin 60°$.
18	A 60.5	The central angle COB is double the inscribed angle: 60°, so the sector is $\frac{60}{360}$ of the circle.
19	E 233	Find BC by the Pythagorean theorem; area = ½·AB·BC.
20	A 12.4	In ABC the altitude from B cuts off triangle ABH similar to ABC; scale its perimeter by EF/BC.
21	B 1555	Triangles ABE and ECD are similar, so CE = AB² ÷ BE; area = ½ · CD · CE.
22	C 230	$AE = \sqrt{(AB^2 + BE^2)}$, then EM and MA from coordinates.
23	B 2320	Common difference from the two terms, then $S_n = n/2 \cdot (2a_1 + (n-1)d)$.
24	A 163	Work done by the pipe alone first; the rest at the net rate 1/F − 1/D.
25	A 24,130	Area of a regular pentagon is $5s^2/(4 \tan 36°)$; multiply by the depth and by 7.48.
26	B 39 ft 3 in	Add the first drop and then a rise and a fall for each bounce: $h + 2h(r + r^2 + \ldots + r^{n-1})$.
27	B −13	b = −k(sum of roots) and c = k(product of roots).
28	B 180.279	Ratio r from $a_k/a_j = r^{k-j}$; then $S_n = a_1(1 - r^n)/(1 - r)$.
29	C $(-\pi/2, \pi/2)$	Tan^{-1} takes the values between its limiting angles; the endpoints are included only if they are attained.
30	E 14.0	Complete the squares; the foci lie on the major axis, symmetric about the center.
31	C 252	(0.7n + 42)/(n + 42) = 0.75; the total is n + 42.
32	E $y = -1\frac{1}{4}$	p = 1/(4a) from a lattice point; directrix is p below the vertex.
33	D 6.7	Locate each focus (vertex plus p), then the distance formula.
34	D 11.33 g	Find the growth or decay factor from the two readings, then run the model back to time 0.
35	C 1359	Use |carrier position − turning point| = j(T − t_1) and solve for the outbound time t_1.
36	C 136	The two courses are perpendicular: $d = \sqrt{(d_1^2 + d_2^2)}$.

#	Answer	Method
37	E 1:41 PM	Distance apart grows at s_A + s_C once both have left; solve s_A(T − t_A) + s_C(T − t_C) = D.
38	C 1.0	Center and a, b from the vertices θ = 0 and θ = π and the focus at the origin; compare the asymptote with the hyperbola at x.
39	C 0.71	Hypergeometric: sum over j ≥ 3 of C(19,j)C(7,4−j) ÷ C(26,4)
40	B 0.8646	Count outcomes (out of $6^5 = 7776$) whose product is a multiple of 4
41	B $\frac{7}{8}$	Focus is p = 1/(4a) from the vertex along the axis; add its coordinates.
42	B $-32\frac{1}{3}$	$f(x_2) = f(x_1)$ + the integral of the parabola from x_1 to x_2.
43	B −4.8	Solve f(x) = c for x numerically, then y-intercept = c − f′(x)·x.
44	B 22.25	Width 0.5; add f at x = 2.5, 3, 3.5, 4 and multiply by 0.5.
45	C $25\frac{1}{3}$	Integrate: 2c − (q³ − p³)/3.
46	D 24g′(15)	Chain rule: f′(x) = g′(p(x))·p′(x); p(3) = 15, p′(3) = 24.
47	C −0.3	Differentiate with respect to t: $F_x \cdot dx/dt + F_y \cdot dy/dt = 0$.
48	C 10.0	$d^2y/dx^2 = [d/dt(dy/dx)] \div (dx/dt)$, with dy/dx = (dy/dt)/(dx/dt).
49	A 26	Signed area: add the trapezoids (areas below the axis count negative).
50	B 9	$\int h' = h(b) - h(a)$, read from the graph.
51	D 0.82	r is symmetric in x and y
52	E 4.24	E(X) = Σ x·P(x)
53	E 0.88	z = (3.5 − 4.2) ÷ 0.6 = −1.17; area above z
54	C 975	Add mean, median and mode
55	A 0	Fences Q1 − 1.5·IQR = 123.25 and Q3 + 1.5·IQR = 577.25
56	D 92nd	z = (80 − 71) ÷ 6.5 = 1.38
57	E 8.77	IQR = 2(0.6745)σ
58	B 118.8	σ = (x − μ) ÷ z with z = 2.0537 for the 98th percentile
59	C 11.8%	Total probability: positive = P(has)P(positive | has) + P(not)P(positive | not)
60	C 24th	z = (3.4 − 3.9) ÷ 0.7 = −0.71; Φ(z)

Practice Test 3 — how each answer is found

#	Answer	Method
1	B 878	Total correct × 6 − total wrong × 2.
2	E $27.23	Add the charges, add the tax, and round to the cent.
3	A −22	Parallel lines share a slope; then b = y − slope · x at the given point.
4	E 10	Perpendicular slopes multiply to −1.
5	C 41.9	Add the three side lengths found with the distance formula.
6	E −21.7	Parallel lines have equal slopes.
7	B 9.2	The midsegment is half the third side AC.
8	B 467 rpm	Inches per minute = mph × 63,360 ÷ 60; divide by the tire circumference $2\pi r$.
9	A 235	Express the second and third counts in the first, then use the given count or total.
10	D 800	Two equations: value and mass; solve the 2×2 system.
11	E 21.9	Central angle $\theta = 2\sin^{-1}(c/2r)$; arc = $r\theta$.
12	B 117°	Opposite angles of an inscribed quadrilateral are supplementary.
13	A 163°	Arc NMW = 2·63°, arc KWM = 2·60°, so arc WK = 2·60 − 49; arc KN = 360 − 2·63 − arc WK.
14	D 141.4	Sector COX has central angle 180 − 108 = 72°: area = $(\frac{72}{360})\pi \cdot 15^2$.
15	B 54.3	Two radii of 15 plus the chord $2 \cdot 15 \cdot \sin(54°)$.
16	B 197	Find the circumradius or side from the given quantity, then apply the area or perimeter formula.
17	A 194	With legs a, b: $m_1^2 = b^2 + a^2/4$ and $m_2^2 = a^2 + b^2/4$. Their sum gives $c^2 = 0.8(m_1^2 + m_2^2)$; then each leg follows.
18	C 167	Right angle at K: legs $2r\cdot\cos 30°$ and $2r\cdot\sin 30°$, so area = $r^2 \sin 60°$.
19	B 101.0	The central angle LOK is double the inscribed angle: 60°, so the sector is $\frac{60}{360}$ of the circle.
20	D 9623	$SA = \pi r^2 + 2\pi r(mr) + 2\pi r^2 = (2m + 3)\pi r^2$ gives r; $V = \pi r^2(mr) + (\frac{2}{3})\pi r^3$.
21	E 9.8%	Geometric mean of the three yearly growth factors, minus 1.
22	C −4.4°	θ = atan2(j, i) (check the quadrant); adding or subtracting 360° gives other valid angles.
23	E 21 yr	Solve (1 + r/12)^(12t) = 3 with logarithms.
24	E −1	Use the given value to find the constant, then solve $f^{-1}(y) = x$ for y.
25	B 174	Find the circumradius or side from the given quantity, then apply the area or perimeter formula.
26	A 165	Find the circumradius or side from the given quantity, then apply the area or perimeter formula.
27	B 28	$d_1 = v\cdot a$, $d_2 = (v + \delta)\cdot b$; the law of cosines gives a quadratic in v.
28	C 871	Distances traveled d = rate × time (convert knots), the angle between the paths from the bearings, then the law of cosines.
29	C −8	Substitute x = −1 and solve for d.
30	C 3	With d known, count the distinct real roots of the quintic.
31	B 26.8	Slope gives b/a (or a/b); then $2c = 2\sqrt{(a^2 + b^2)}$.
32	A −8.0	It lies on the perpendicular bisector of PQ: use the midpoint and slope $-\Delta x/\Delta y$.
33	D 28.0	$T = \text{mid} - \text{amp}\cdot\cos(2\pi t/P)$; $T \geq$ thr when cos is small enough: time = $(P/\pi)\cdot\arccos((\text{thr} - \text{mid})/\text{amp})$.
34	C 44.2	Three-dimensional distance formula for each side, then add.
35	A 172	Solve y(t) = 0 for the positive t, then x = (v cos θ)t.
36	A 0.3	t = ln x − 3, so y = 3(x/e^3)^2.

#	Answer	Method
37	B 3078	A senior for the first office, then ordered choices from everyone left for the other two.
38	A 3	A rose r = a cos nθ has n petals when n is odd and 2n petals when n is even.
39	E −5	Vertex form $y = a(x - h)^2 + k$; focus is $(h, k + 1/(4a))$.
40	E 3.9	Substitute the line into the parabola; distance = $\sqrt{(1+m^2)} \cdot \sqrt{(\text{disc})}/|a|$.
41	A 2.75	a from P or Q, $p = 1/(4a)$; y-coordinate $k + p$.
42	A −6.0	$f'(x) = 2a(x - h)$.
43	B 10.7	Area = $\lvert a \rvert (x_2 - x_1)^3/6$.
44	C 39.8 m	Rectangle area = $2r \cdot L$ with $2L + 2\pi r = P$; maximize: $r = P/4\pi$.
45	E $-\frac{1}{7}$	$f(-2) = 7$, so $g'(7) = 1/f'(-2) = 1/(-7)$.
46	B 6	f can only change direction where f′ changes sign: compare f at the endpoints and those points, using areas.
47	B −40.83	f can only change direction where f′ changes sign: compare f at the endpoints and those points, using areas.
48	C 3.24	Step size 0.3: $y_1 = y_0 + h \cdot f(0, y_0)$, $y_2 = y_1 + h \cdot f(0.3, y_1)$.
49	C 3.72	Separate variables: $\ln y = (c/3)x^3 + C$, so $y = y_0 e^{(cx^3/3)}$.
50	C −0.50	Implicit differentiation: $dy/dx = -F_x/F_y$; solve for y first.
51	B $0.\overline{1}$	P(A | B) = P(A and B) ÷ P(B)
52	B 1.66	$P_3(x) = f(a) + f'(a)(x-a) + f''(a)(x-a)^2/2! + f'''(a)(x-a)^3/3!$.
53	E 0.063	$P = 2 \cdot P(Z > \lvert z \rvert)$
54	A 7.0	b = Sxy ÷ Sxx
55	D −6.3	Residual = $y - \hat{y}$, with $\hat{y}$ from the least-squares line ($y = a + bx$)
56	A 0.23	Total has mean 110.7 and sd $\sqrt{(\Sigma\sigma^2)} = 2.528$
57	C 4.49	$E(X) = \Sigma\, x \cdot P(x)$
58	A 0.040	$\chi^2 = \Sigma(O - E)^2 \div E = 10.053$, df = 4
59	B 1.68	$(O - E)^2 \div E$ with E = 38.00
60	D 3	Score = 126 + 2.5·60 = 276; 6(A − w) − 2w = 276 with A = 50, so w = (6A − 276) ÷ 8